TTG application performance

30 entries: one microbenchmark plus29 full applications; all4 configurations.

Actual driver/socket observations, complete domain workflows, native correctness and restart checks.

Seed42; >=3 independent measured primary trials per cell; one excluded workflow warmup.

Primary protocol time, driver time and SQL/control counts stay separate. Clean HTTP controls are deferred.

Measurement boundaries

Protocol completion wall time: prebuilt packet -> real sendall -> PostgreSQL/transport/decoding -> ReadyForQuery Z.

Driver-call time separately includes parameter encoding. Compiler/SQL construction/model CPU is outside protocol time.

Count actual SQL dispatches, Sync/Z cycles, CommandComplete C and DataRow D; retain controls and expected errors.

Natural close remains unchanged; late workers are drained and traced after the app response.

Observed outcomes

Legacy TTG: lower full-workflow protocol median for LinkedList; higher medians for all29 full applications in this local fixed-seed study.

Both dependency policies: unchanged SQL counts versus none for every application.

All60 dependency preflight cases considered zero plans. Actual checkpoint barriers remain visible; this is zero observed mechanism coverage.

n=3 with dispersion reported. Timing differences without mechanism activity are not evidence of dependency-prefetch benefit.

Experimental controls

Fresh equivalent DB each trial, seeded/verified pages warm; fresh session per request; compiler/OS warm. Module SQL-template caches not forcibly reset beyond process lifecycle; setup may warm them.

Cold provider reset on first workflow request; adaptive within workflow. Three warmup final-state reads then three steady reads. Profiles discarded with process; no held-out training.

Fresh equivalent state and deterministic actions; rotated policy order; no held-out training.

Clean app HTTP/e2e is deferred: the frozen clean path does not honor the cold-provider reset. No clean times are substituted from instrumented HTTP.

SQL-only calibration

Observed deltas and reference variation; no fixed subtraction

Synthetic SQL	Outer delta (us)	Protocol delta (us)	Reference medians range (us)
scalar	+2.240	-1.489	39.65–41.76
text16k	+2.022	+2.050	149.58–151.63
json16k	+2.305	-3.043	165.02–165.85
multirow128	+9.872	-3.717	315.57–318.84

Rotated repeats; statement logging off for controls. Independent PG CSV checks and profiles are separate diagnostics.

Full ordered workflows: applications 1–10

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
LinkedList	93.03 [91.71, 93.51]	80.01 [79.08, 80.28]	122.93 [109.94, 123.55]	98.01 [94.99, 100.37]
LittleX	42.40 [41.65, 44.12]	121.95 [114.29, 126.84]	40.44 [39.67, 47.61]	43.31 [42.79, 49.72]
Jacord	119.16 [118.98, 119.50]	129.36 [129.04, 129.82]	119.88 [118.00, 126.20]	117.35 [117.05, 118.47]
JDrive	47.82 [47.32, 48.60]	78.01 [76.94, 80.91]	48.03 [47.81, 49.75]	79.55 [64.27, 675.88]
IssueDesk	142.43 [140.88, 146.30]	301.22 [298.36, 302.47]	144.61 [144.11, 147.52]	140.29 [139.85, 142.29]
ApprovalDesk	131.32 [131.31, 131.86]	279.36 [273.68, 280.60]	130.23 [128.59, 131.05]	130.44 [129.09, 131.72]
StockRoom	136.10 [134.09, 136.16]	325.91 [324.38, 329.18]	136.57 [135.41, 138.15]	138.11 [135.91, 139.84]
ClinicQueue	107.74 [107.17, 108.60]	213.78 [208.03, 216.46]	106.82 [104.42, 107.58]	104.49 [103.59, 105.16]
CoursePath	158.73 [156.65, 165.41]	272.77 [272.73, 273.11]	159.52 [158.17, 165.15]	156.26 [154.23, 162.79]
LibraryLoans	145.60 [142.78, 146.50]	308.94 [308.18, 323.01]	148.69 [146.75, 148.82]	146.02 [143.46, 146.04]

Full ordered workflows: applications 11–20

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
EventSeats	105.74 [105.18, 116.52]	218.21 [217.84, 250.85]	108.20 [104.81, 108.56]	104.43 [104.33, 106.75]
HelpQueue	161.24 [160.82, 171.87]	327.05 [324.09, 332.38]	159.93 [155.25, 160.04]	157.39 [156.83, 158.46]
RecipePantry	86.34 [83.69, 86.58]	198.61 [197.46, 198.90]	87.75 [84.06, 90.83]	91.30 [84.28, 98.90]
TripBoard	48.75 [48.67, 50.24]	122.07 [121.80, 122.94]	49.22 [47.76, 49.23]	48.36 [47.56, 52.24]
DeliveryRoute	99.44 [98.97, 102.96]	229.54 [228.74, 234.14]	109.48 [104.91, 110.10]	106.45 [103.11, 107.29]
HiringPipeline	192.38 [189.54, 193.43]	387.24 [385.76, 389.04]	180.15 [178.08, 180.19]	185.77 [180.17, 185.86]
SalesPipeline	64.50 [63.61, 64.66]	153.47 [153.20, 156.55]	65.10 [64.80, 65.61]	64.93 [64.69, 64.94]
GrantReview	78.25 [71.36, 78.50]	167.98 [167.80, 170.25]	68.67 [68.12, 68.89]	77.09 [75.19, 78.66]
LabSamples	77.89 [75.72, 78.24]	191.11 [190.52, 191.28]	75.11 [75.04, 75.88]	79.02 [78.30, 79.39]
AssetCare	61.67 [60.07, 62.11]	162.25 [160.71, 162.34]	61.67 [60.89, 62.92]	62.04 [61.55, 62.93]

Full ordered workflows: applications 21–30

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
UtilityMeter	62.38 [58.88, 63.04]	131.34 [130.81, 134.31]	59.42 [56.13, 63.41]	63.94 [61.18, 69.05]
VolunteerShift	76.11 [76.01, 76.86]	159.16 [155.79, 165.55]	76.27 [74.04, 76.35]	75.58 [75.11, 76.42]
PollAssembly	65.82 [64.66, 68.84]	147.31 [145.47, 147.50]	64.49 [63.34, 65.90]	69.53 [66.67, 71.39]
BudgetLedger	57.85 [54.65, 66.49]	137.62 [135.83, 137.86]	53.89 [52.57, 56.17]	60.03 [56.16, 69.73]
BuildRelease	92.77 [92.18, 93.13]	201.46 [199.32, 203.96]	91.65 [91.57, 91.74]	93.31 [90.36, 93.91]
KnowledgeMap	74.00 [72.89, 97.59]	170.71 [168.96, 218.50]	71.94 [70.83, 72.42]	72.96 [70.88, 74.87]
ReturnCounter	57.83 [57.50, 63.49]	135.07 [134.60, 135.22]	57.27 [56.68, 58.58]	55.43 [54.73, 56.14]
GardenPlan	67.18 [65.90, 68.95]	154.78 [153.86, 155.69]	63.33 [63.08, 66.03]	63.19 [62.20, 66.40]
Tournament	72.32 [70.19, 72.32]	131.32 [130.46, 131.39]	72.12 [71.23, 75.47]	71.17 [69.86, 72.39]
IncidentRoom	77.00 [71.78, 79.18]	190.21 [188.95, 191.54]	77.57 [73.55, 94.41]	67.40 [66.02, 67.91]

First workflow request: cold model: applications 1–10

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
LinkedList	21.23 [20.74, 21.28]	16.12 [15.32, 17.25]	50.77 [36.28, 51.54]	22.68 [21.77, 26.85]
LittleX	5.19 [5.18, 5.34]	22.92 [21.83, 23.05]	5.13 [4.99, 5.24]	5.18 [5.11, 5.33]
Jacord	47.45 [47.13, 47.92]	48.12 [47.82, 48.44]	48.28 [47.73, 52.08]	47.30 [46.99, 47.40]
JDrive	3.86 [3.74, 3.88]	3.91 [3.77, 4.00]	3.53 [3.47, 3.59]	3.77 [3.61, 3.84]
IssueDesk	6.78 [6.77, 6.81]	16.83 [16.64, 17.02]	7.02 [6.95, 7.06]	6.97 [6.71, 7.03]
ApprovalDesk	4.13 [4.00, 4.17]	10.33 [10.18, 10.48]	4.07 [3.89, 4.20]	4.11 [3.81, 4.18]
StockRoom	6.04 [6.01, 6.35]	24.02 [23.91, 26.72]	6.19 [6.10, 6.20]	5.93 [5.92, 6.07]
ClinicQueue	9.58 [9.40, 9.65]	21.30 [21.26, 21.78]	9.49 [9.02, 11.08]	9.32 [9.23, 9.37]
CoursePath	7.10 [6.82, 7.14]	17.71 [17.43, 17.88]	7.46 [7.14, 7.48]	6.85 [6.81, 7.04]
LibraryLoans	7.37 [7.26, 7.40]	25.32 [25.32, 25.38]	7.22 [7.18, 7.30]	6.79 [6.68, 6.83]

First workflow request: cold model: applications 11–20

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
EventSeats	6.83 [6.75, 6.86]	17.95 [17.70, 17.96]	6.56 [6.55, 6.76]	7.01 [6.90, 7.15]
HelpQueue	6.68 [6.40, 6.79]	17.55 [17.46, 17.67]	6.47 [6.39, 6.49]	6.52 [6.48, 6.62]
RecipePantry	6.75 [6.43, 6.82]	20.50 [20.22, 20.54]	6.20 [6.14, 6.52]	6.62 [6.34, 6.68]
TripBoard	3.83 [3.79, 3.92]	12.63 [12.63, 12.75]	3.68 [3.67, 3.87]	3.68 [3.66, 3.86]
DeliveryRoute	5.27 [5.22, 5.44]	19.04 [18.83, 19.20]	5.35 [5.20, 5.49]	5.48 [5.26, 5.49]
HiringPipeline	3.75 [3.75, 3.94]	14.23 [14.19, 14.43]	3.54 [3.52, 3.67]	3.87 [3.84, 3.99]
SalesPipeline	5.95 [5.68, 5.99]	15.07 [14.84, 15.10]	5.83 [5.77, 5.84]	5.71 [5.64, 5.79]
GrantReview	5.06 [4.96, 5.19]	18.03 [17.55, 18.06]	4.99 [4.96, 5.25]	5.41 [5.30, 5.58]
LabSamples	4.76 [4.73, 4.79]	17.57 [17.42, 17.63]	4.43 [4.39, 4.45]	4.48 [4.41, 4.51]
AssetCare	5.72 [5.44, 5.80]	18.49 [18.45, 19.05]	5.58 [5.51, 5.60]	5.79 [5.73, 5.81]

First workflow request: cold model: applications 21–30

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
UtilityMeter	4.30 [4.09, 4.32]	15.75 [15.64, 15.90]	3.87 [3.85, 3.97]	4.06 [4.00, 4.22]
VolunteerShift	7.27 [7.26, 7.56]	19.36 [19.34, 19.52]	7.59 [7.28, 7.87]	7.41 [7.37, 7.49]
PollAssembly	6.75 [6.19, 6.89]	18.58 [18.38, 18.62]	6.06 [5.91, 6.11]	6.72 [6.21, 6.79]
BudgetLedger	5.30 [5.01, 5.39]	18.63 [18.47, 18.69]	5.29 [5.20, 5.30]	5.26 [5.25, 5.34]
BuildRelease	2.45 [2.39, 2.46]	10.59 [10.53, 10.67]	2.36 [2.31, 2.39]	2.43 [2.40, 2.44]
KnowledgeMap	9.20 [8.87, 9.33]	21.55 [21.20, 21.62]	9.22 [9.04, 9.36]	9.16 [8.84, 9.32]
ReturnCounter	3.77 [3.64, 3.87]	12.52 [12.44, 12.59]	3.73 [3.68, 3.79]	3.67 [3.58, 3.69]
GardenPlan	3.84 [3.77, 3.89]	14.18 [14.14, 14.23]	3.68 [3.62, 3.90]	3.86 [3.62, 3.97]
Tournament	9.97 [9.69, 9.97]	19.29 [18.96, 19.35]	9.99 [9.94, 10.09]	9.96 [9.86, 10.27]
IncidentRoom	3.96 [3.90, 3.99]	18.70 [18.69, 18.77]	4.20 [4.06, 4.21]	3.91 [3.83, 3.97]

Final-state read-only phase: applications 1–10

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
LinkedList	57.56 [57.09, 58.83]	31.52 [30.26, 32.09]	62.02 [60.98, 62.57]	60.80 [60.67, 61.60]
LittleX	15.47 [15.17, 16.16]	52.77 [46.77, 52.92]	15.09 [14.95, 15.39]	14.41 [13.97, 14.68]
Jacord	147.20 [146.56, 147.88]	133.19 [132.24, 133.28]	147.10 [144.51, 147.76]	147.46 [143.76, 148.43]
JDrive	29.61 [28.48, 29.71]	53.66 [49.76, 54.60]	30.31 [29.84, 30.67]	30.72 [29.55, 42.86]
IssueDesk	25.17 [24.73, 25.74]	44.68 [44.41, 44.97]	25.56 [24.99, 26.40]	25.03 [24.81, 26.25]
ApprovalDesk	33.46 [33.06, 33.72]	50.33 [50.30, 50.65]	33.46 [32.22, 33.71]	33.20 [32.56, 33.28]
StockRoom	38.93 [37.47, 38.96]	73.22 [72.75, 73.37]	38.97 [37.84, 39.09]	38.68 [37.53, 39.10]
ClinicQueue	34.01 [33.53, 34.11]	50.56 [50.09, 50.58]	33.43 [33.11, 33.79]	33.09 [33.02, 33.13]
CoursePath	41.79 [40.63, 41.81]	59.25 [59.19, 60.38]	40.94 [40.49, 42.52]	38.21 [38.20, 39.55]
LibraryLoans	39.38 [39.09, 39.69]	76.21 [75.99, 80.76]	40.06 [40.00, 40.22]	39.56 [39.22, 39.86]

Three measured reads after three warmup reads in each fresh trial; final mutated dataset, warmed profile.

Final-state read-only phase: applications 11–20

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
EventSeats	33.87 [33.63, 36.60]	50.62 [49.94, 50.71]	34.27 [32.97, 35.61]	33.66 [33.56, 33.84]
HelpQueue	35.53 [34.97, 35.56]	57.08 [56.88, 57.55]	34.42 [34.17, 35.18]	35.61 [34.46, 35.74]
RecipePantry	24.20 [23.42, 25.00]	41.16 [40.66, 41.42]	26.81 [26.30, 26.87]	26.07 [24.28, 26.47]
TripBoard	13.59 [12.91, 13.93]	30.26 [29.81, 30.62]	12.21 [11.97, 12.62]	12.38 [12.18, 12.67]
DeliveryRoute	24.43 [22.88, 25.02]	46.19 [46.00, 46.25]	24.06 [23.71, 24.93]	23.24 [22.56, 23.91]
HiringPipeline	36.99 [36.93, 38.92]	57.04 [56.93, 57.49]	38.20 [38.12, 39.59]	40.97 [39.51, 41.51]
SalesPipeline	17.79 [17.64, 18.13]	34.78 [34.48, 35.28]	18.32 [18.32, 18.41]	17.93 [17.57, 18.01]
GrantReview	23.06 [22.72, 23.53]	48.36 [48.29, 48.52]	26.14 [24.10, 29.24]	25.14 [24.59, 25.80]
LabSamples	21.31 [20.87, 21.52]	46.65 [46.09, 46.70]	21.17 [20.70, 21.31]	21.04 [21.01, 21.57]
AssetCare	22.77 [21.81, 22.89]	43.13 [43.06, 43.58]	21.02 [20.79, 22.30]	20.91 [20.85, 21.40]

Three measured reads after three warmup reads in each fresh trial; final mutated dataset, warmed profile.

Final-state read-only phase: applications 21–30

Each cell: median [Q1, Q3] summed protocol milliseconds. n=3 independent trials; quartiles are not confidence intervals.

Application	None	Original TTG	Dep. always	Dep. benefit
UtilityMeter	11.99 [11.99, 12.08]	32.32 [32.25, 32.72]	11.77 [11.61, 11.87]	12.27 [11.74, 12.73]
VolunteerShift	26.80 [26.52, 27.14]	49.22 [48.31, 49.45]	25.83 [25.79, 26.71]	26.62 [26.61, 26.72]
PollAssembly	21.00 [20.98, 22.55]	43.00 [42.85, 43.40]	23.16 [22.39, 23.80]	23.36 [22.48, 23.88]
BudgetLedger	16.97 [16.82, 17.97]	38.39 [38.02, 38.74]	17.11 [17.01, 17.13]	19.42 [17.70, 19.70]
BuildRelease	19.45 [19.41, 19.74]	37.35 [36.81, 37.43]	19.09 [19.08, 19.14]	19.31 [18.56, 19.42]
KnowledgeMap	27.29 [26.73, 27.48]	49.36 [49.21, 49.87]	26.26 [26.14, 26.34]	26.93 [26.29, 26.95]
ReturnCounter	16.03 [15.74, 16.53]	32.87 [32.65, 33.11]	15.26 [14.74, 15.37]	16.02 [15.53, 16.73]
GardenPlan	23.67 [23.33, 24.98]	41.59 [41.57, 41.78]	24.28 [23.12, 24.39]	21.54 [21.27, 23.25]
Tournament	31.78 [31.09, 32.05]	50.66 [49.70, 51.12]	31.67 [31.18, 31.87]	33.01 [32.73, 33.02]
IncidentRoom	23.10 [21.62, 23.28]	47.60 [47.15, 47.91]	24.23 [23.32, 24.46]	23.04 [22.16, 23.05]

Three measured reads after three warmup reads in each fresh trial; final mutated dataset, warmed profile.

Actual SQL counts: applications 1–10

Full workflow: total / non-control data / control dispatches. Each count is constant across all three measured trials.

Application	None	Original TTG	Dep. always	Dep. benefit
LinkedList	355 / 313 / 42	105 / 51 / 54	355 / 313 / 42	355 / 313 / 42
LittleX	153 / 90 / 63	154 / 79 / 75	153 / 90 / 63	153 / 90 / 63
Jacord	293 / 265 / 28	231 / 196 / 35	293 / 265 / 28	293 / 265 / 28
JDrive	156 / 107 / 49	135 / 80 / 55	156 / 107 / 49	156 / 107 / 49
IssueDesk	555 / 366 / 189	577 / 358 / 219	555 / 366 / 189	555 / 366 / 189
ApprovalDesk	539 / 343 / 196	577 / 341 / 236	539 / 343 / 196	539 / 343 / 196
StockRoom	497 / 343 / 154	493 / 312 / 181	497 / 343 / 154	497 / 343 / 154
ClinicQueue	408 / 275 / 133	403 / 250 / 153	408 / 275 / 133	408 / 275 / 133
CoursePath	521 / 381 / 140	544 / 376 / 168	521 / 381 / 140	521 / 381 / 140
LibraryLoans	531 / 370 / 161	522 / 336 / 186	531 / 370 / 161	531 / 370 / 161

Actual SQL counts: applications 11–20

Full workflow: total / non-control data / control dispatches. Each count is constant across all three measured trials.

Application	None	Original TTG	Dep. always	Dep. benefit
EventSeats	434 / 294 / 140	436 / 274 / 162	434 / 294 / 140	434 / 294 / 140
HelpQueue	605 / 409 / 196	607 / 377 / 230	605 / 409 / 196	605 / 409 / 196
RecipePantry	327 / 208 / 119	335 / 186 / 149	327 / 208 / 119	327 / 208 / 119
TripBoard	232 / 134 / 98	246 / 133 / 113	232 / 134 / 98	232 / 134 / 98
DeliveryRoute	415 / 268 / 147	416 / 246 / 170	415 / 268 / 147	415 / 268 / 147
HiringPipeline	753 / 494 / 259	806 / 477 / 329	753 / 494 / 259	753 / 494 / 259
SalesPipeline	292 / 180 / 112	307 / 167 / 140	292 / 180 / 112	292 / 180 / 112
GrantReview	300 / 188 / 112	310 / 170 / 140	300 / 188 / 112	300 / 188 / 112
LabSamples	338 / 212 / 126	357 / 199 / 158	338 / 212 / 126	338 / 212 / 126
AssetCare	273 / 168 / 105	276 / 145 / 131	273 / 168 / 105	273 / 168 / 105

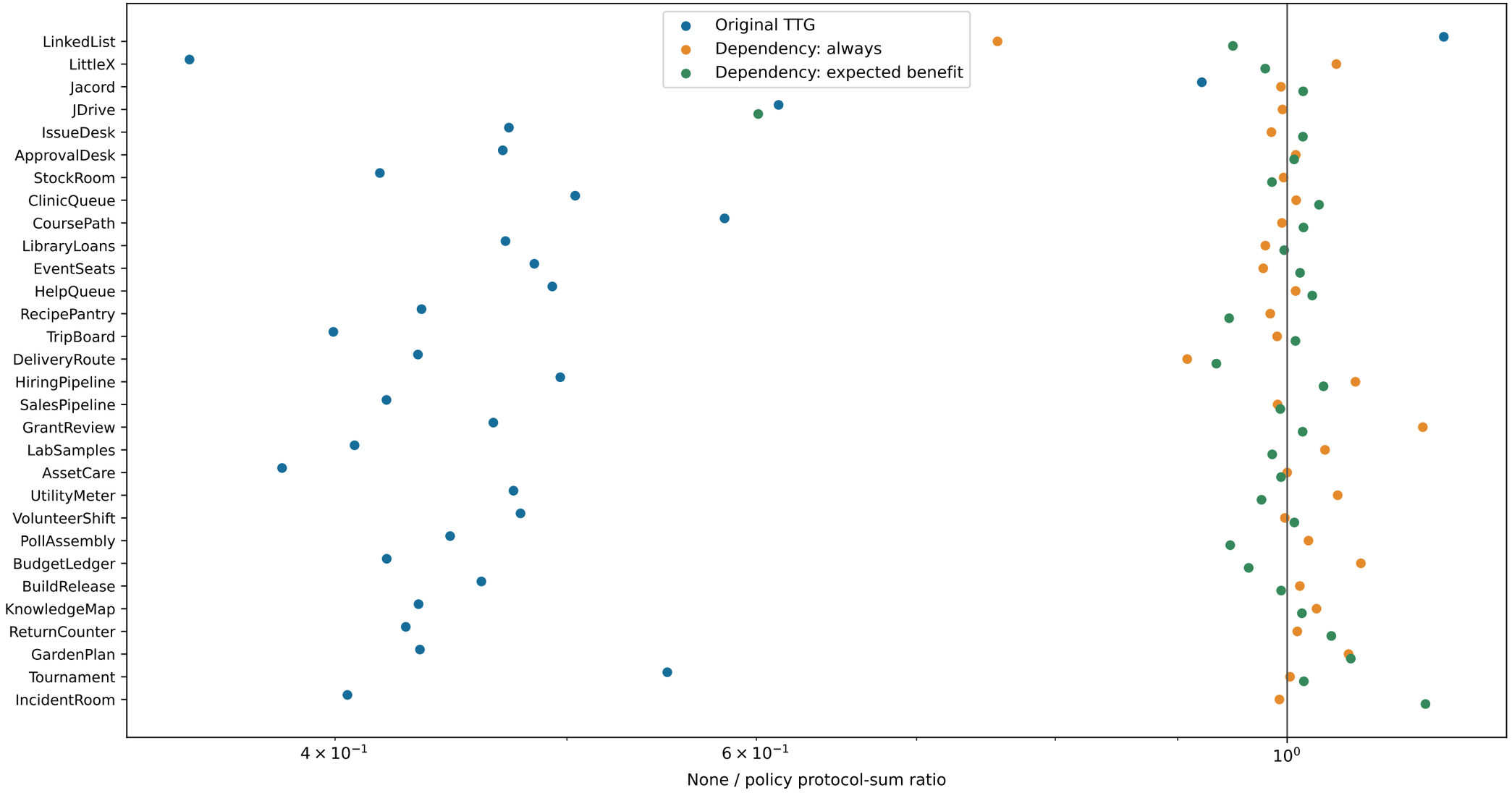
Actual SQL counts: applications 21–30

Full workflow: total / non-control data / control dispatches. Each count is constant across all three measured trials.

Application	None	Original TTG	Dep. always	Dep. benefit
UtilityMeter	244 / 153 / 91	249 / 144 / 105	244 / 153 / 91	244 / 153 / 91
VolunteerShift	286 / 195 / 91	279 / 174 / 105	286 / 195 / 91	286 / 195 / 91
PollAssembly	270 / 179 / 91	263 / 158 / 105	270 / 179 / 91	270 / 179 / 91
BudgetLedger	234 / 143 / 91	235 / 130 / 105	234 / 143 / 91	234 / 143 / 91
BuildRelease	390 / 243 / 147	411 / 242 / 169	390 / 243 / 147	390 / 243 / 147
KnowledgeMap	298 / 186 / 112	304 / 164 / 140	298 / 186 / 112	298 / 186 / 112
ReturnCounter	259 / 154 / 105	274 / 153 / 121	259 / 154 / 105	259 / 154 / 105
GardenPlan	286 / 181 / 105	294 / 163 / 131	286 / 181 / 105	286 / 181 / 105
Tournament	265 / 188 / 77	276 / 187 / 89	265 / 188 / 77	265 / 188 / 77
IncidentRoom	293 / 174 / 119	305 / 166 / 139	293 / 174 / 119	293 / 174 / 119

Per-application relative protocol sums

Ratio of measured medians: none / policy. Full dispersion remains in CSV/JSON; small differences are uncertain.



Diagnostic mechanism coverage: 1–10

Observed considered dependency plans in correctness preflight; zero is retained, not a successful-prefetch claim.

Application	Always considered	Expected-benefit considered
LinkedList	0	0
LittleX	0	0
Jacord	0	0
JDrive	0	0
IssueDesk	0	0
ApprovalDesk	0	0
StockRoom	0	0
ClinicQueue	0	0
CoursePath	0	0
LibraryLoans	0	0

Diagnostic mechanism coverage: 11–20

Observed considered dependency plans in correctness preflight; zero is retained, not a successful-prefetch claim.

Application	Always considered	Expected-benefit considered
EventSeats	0	0
HelpQueue	0	0
RecipePantry	0	0
TripBoard	0	0
DeliveryRoute	0	0
HiringPipeline	0	0
SalesPipeline	0	0
GrantReview	0	0
LabSamples	0	0
AssetCare	0	0

Diagnostic mechanism coverage: 21–30

Observed considered dependency plans in correctness preflight; zero is retained, not a successful-prefetch claim.

Application	Always considered	Expected-benefit considered
UtilityMeter	0	0
VolunteerShift	0	0
PollAssembly	0	0
BudgetLedger	0	0
BuildRelease	0	0
KnowledgeMap	0	0
ReturnCounter	0	0
GardenPlan	0	0
Tournament	0	0
IncidentRoom	0	0

LittleX: measured recursive SQL contributions

Primary SQL traces, n=3. Mean per-trial fingerprint contributions; distinct from workload median differences.

Recursive fingerprint SHA prefix	Mean dispatches: none -> TTG	Protocol-sum delta (ms)
d36899011a19	0 -> 4	+57.048
ebc1e8b69ff7	0 -> 1	+11.431
5c85d1cc2354	0 -> 1	+6.215
5644fab3cc55	0 -> 1	+4.873

Workflow medians: 42.405 -> 121.952 ms. SQL dispatches: 153 -> 154; controls: 63 -> 75.

Four added recursive fingerprints account for observed SQL contributions, alongside fewer other queries.

Full fingerprints and three-trial values are in accepted-db-observations.json; no CPU cause is inferred.

Reached checkpoints: precise admission barriers

LittleX load_feed.run: checkpoint at server.jac:265, Profile.collect target; stable=true, path_determined=false, execution_preserved=true; status=blocked.

Its prediction reports "Unknown effect or cancellation at 273". Compiler event5 is a barrier for collect at line273: the Follow/Profile/Post/Tweet multi-hop visit.

Jacord gather_messages: checkpoint at server.impl.jac:242 targets Message.collect_message; blocked because prediction reports unknown effect at246, where report _message_view(here) occurs.

Line numbers refer to sealed executed preflight copies, including the Jacord adapter prefix. These are reached checkpoints with predicted barriers, not proof that future statements executed.

Separate captures: checkpoint activity and considered plans

Actual preflight versus new detailed diagnostics. Each entry is checkpoint hits / considered plans; no pooling.

Application	Policy	Preflight	New diagnostic
ApprovalDesk	always	15 / 0	15 / 0
ApprovalDesk	expected benefit	15 / 0	15 / 0
AssetCare	always	33 / 0	33 / 0
AssetCare	expected benefit	33 / 0	33 / 0
HiringPipeline	always	22 / 0	22 / 0
HiringPipeline	expected benefit	22 / 0	22 / 0
LittleX	always	22 / 0	22 / 0
LittleX	expected benefit	22 / 0	22 / 0
TripBoard	always	11 / 0	11 / 0
TripBoard	expected benefit	11 / 0	11 / 0

Validated diagnostics and excluded CPU capture

20 representative app/policy diagnostic cases reconcile driver dispatches/errors with PostgreSQL CSV logs.

15,413 driver dispatches = 15,373 PG execute records +40 matching expected SQL-error records; strict per-fingerprint checks passed.

CPU capture was unsuccessful: 167/1432 profiles have negative self/cumulative entries. All CPU leaders/comparisons are excluded pending controlled validation.

Raw profiles and the original checker failure are preserved. Primary SQL timing/counts and actual checkpoint decisions provide the accepted explanation evidence.

Interpretation and reproducibility

This is a local seed42 smoke/domain study. No held-out seeds or injected-latency result is implied.

Summed DB intervals may overlap; interval unions and foreground/background counts are separate.

Unknown payload consumption bytes and pure blocked wait stay unknown. Demand predictions are not payload promotions.

Full machine tables, raw timestamps/SQL, action/data manifests, source hashes, profiles and main review artifacts accompany this deck.